pKa – Viva Questions

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1. What is pKa? Give cell representation of the electrochemical cell set up in the estimation of pKa.

pKa is the negative logarithm (base 10) of the dissociation constant of a weak acid: pKa = −log K_a.

A smaller pKa means a stronger acid.

Cell used (glass electrode as indicator, calomel electrode as reference):

Ag | AgCl(s) | HCl (0.1 M) | glass | CH_3COOH (test solution) || KCl (sat.) | Hg_2Cl_2(s) | Hg

2. Which is the indicator electrode used in the estimation of pKa? Why is it called so?

The glass electrode. It is called an indicator electrode because its potential depends on (indicates) the H^+ ion concentration, i.e. the pH, of the solution in which it is dipped.

3. Name two electrodes required to determine pH of any solution.

Glass electrode (indicator electrode) and saturated calomel electrode (reference electrode). Usually both are built into a single combined glass electrode.

4. How do you relate pH and pKa?

By the Henderson–Hasselbalch equation:

pH = pKa + log ([Salt] / [Acid])

At the half equivalence point [Salt] = [Acid], so log 1 = 0 and pH = pKa.

5. What is dissociation constant?

It is the equilibrium constant for the dissociation of a weak electrolyte into its ions. For acetic acid, $CH_3COOH \rightleftharpoons CH_3COO^- + H^+$:

$K_a = [H^+][CH_3COO^-] / [CH_3COOH]$

It measures the strength of the acid (its tendency to donate a proton).

6. What is glass electrode? Write the cell representation of glass electrode.

A glass electrode is an ion-selective electrode made of a thin bulb of special low-melting, H^+-sensitive glass, containing 0.1 M HCl with a Ag/AgCl wire dipped in it. A potential develops across the glass membrane that depends on the H^+ concentration of the outside solution, so it is used to measure pH.

Ag | AgCl(s) | HCl (0.1 M) | glass

7. What is buffer solution? Name the buffer formed during pKa experiment.

A buffer solution is a solution that resists change in its pH on addition of a small amount of acid or base or on dilution. During the pKa experiment an acidic buffer of acetic acid and sodium acetate (CH_3COOH + CH_3COONa) is formed before the equivalence point.

8. Give an example each for weak acid, strong acid, weak base and strong base.

Weak acid – CH_3COOH (acetic acid)

Strong acid – HCl (hydrochloric acid)

Weak base – NH_4OH (ammonium hydroxide)

Strong base – NaOH (sodium hydroxide)

9. Account for the sudden jump in pH at equivalence point.

Before the equivalence point the acetic acid–sodium acetate buffer keeps the pH nearly constant. At the equivalence point all the acid is neutralised and the buffer is used up, so even a drop of extra NaOH adds free OH^- ions to the solution. This causes a sudden, large increase in pH.

10. What do you mean by equivalence and half equivalence point in the pKa experiment?

Equivalence point: the point at which the amount of NaOH added is chemically equivalent to the acetic acid present, i.e. all the acid is just neutralised. It is shown by the sudden jump in pH (peak in the $\Delta pH/\Delta V$ graph).

Half equivalence point: the point at which exactly half of the acid has been neutralised, i.e. half the equivalence volume. Here $[CH_3COONa] = [CH_3COOH]$, so pH = pKa.

11. Why is pKa not determined for strong acid?

A strong acid is completely dissociated in water, so there is no equilibrium between the undissociated acid and its ions. Its K_a is very large (practically infinite), so pKa has no meaningful value and no buffer is formed during its titration.

12. List the chemicals, apparatus and equipment used in the pKa experiment.

- Chemicals: weak acid (acetic acid) solution, standard NaOH solution, distilled water, standard buffer solutions (for calibrating the pH meter).
- Apparatus: burette, pipette, beaker, glass rod / stirrer, wash bottle.
- Equipment: pH meter with combined glass electrode.

13. Write the nature of the graphs obtained in the pKa experiment.

- pH vs volume of NaOH: an S-shaped curve. pH rises slowly in the buffer region, rises sharply at the equivalence point, and then levels off.
- $\Delta pH/\Delta V$ vs volume of NaOH: a sharp peak. The volume at the peak gives the equivalence point.

14. What are the advantages of pH-metric titration over volumetric analysis?

- No indicator is needed.
- Can be used for coloured, turbid or very dilute solutions.
- The end point is found accurately from the graph, not by eye.
- Besides the end point, it also gives the pKa of the weak acid.

15. To which type of chemical titration does the pH-metric determination of pKa belong?

It is an acid–base (neutralisation) titration – a weak acid (acetic acid) titrated against a strong base (NaOH).